Homework 7: Projections; Regression using Linear Algebra

EECS 245, Fall 2026 at the University of Michigan

due Sunday, November 8th, 2026 at 11:59PM Ann Arbor Time

Except for the video link in Problem 1 and the notebook submission in Problem 8, write your solutions to the following problems either by writing them on a piece of paper or on a tablet and scanning your answers as a PDF. Note that you are not allowed to use LaTeX, Google Docs, or any other digital document creation software to type your answers. Homeworks are due to Pensive by 11:59PM on the due date. See the **syllabus** for details on the slip day policy.

Homework will be evaluated not only on the correctness of your answers, but on your ability to present your ideas clearly and logically. You should always explain and justify your conclusions, using sound reasoning. Your goal should be to convince the reader of your assertions. If a question does not require explanation, it will be explicitly stated.

Before proceeding, make sure you're familiar with the **collaboration policy**.

Total Points: $10 + 5 + 15 + 16 + 11 + 8 + 22 + 13 = 100$

Problem 1: Homework 6 Concept Video (10 pts)

Record a **2- to 5-minute video** answering **one** of the prompts below. These prompts are related to the ideas in **Homework 6**.

1. If you append a copy of one of a matrix's columns to that matrix, what happens to its rank and the dimension of its null space, and why?
2. Suppose a square matrix A maps two different vectors to the same vector. Can A be invertible? Explain how this relates to its null space.
3. Suppose A has more columns than rows. Explain why $A\vec{x} = \vec{b}$ cannot have exactly one solution. Can it have no solutions?
4. Can XX^T be invertible when X has more rows than columns? Explain using rank or linear independence.

The video should be **just you speaking into the camera**. Your face must be visible while you are talking for the entire video. Do not include slides, a screen recording, written work, or any other visuals.

Submit your video link to the separate assignment on Pensive called **Homework 7, Problem 1 Video Link**. You can upload your video to YouTube as unlisted, share a Zoom recording link, or share a Google Drive link; whatever works. Make sure the course staff can view the video using that link.

Your submission time for Homework 7 is the latest of your three submission times: your Problem 1 video link, your PDF, and your code.

This problem is graded on completion. Record a video of the required length and make an honest effort to explain the question, and you will receive full credit. Your explanation does not need to be polished or perfect.

Problem 2: Anonymous Feedback (5 pts)

We'd like to get your feedback on how the course has been going so far, now that we're past the halfway point and Midterm 2 is fast approaching.

You can find the survey **at this link**, which you should complete **after you've finished the rest of Homework 7**. Unlike the Homework 3 survey, **it is anonymous**, so feel free to provide candid feedback.

In order to earn the 5 points for Homework 7, Problem 2, include a screenshot of the confirmation message you see after submitting the form. (We consider it an honor code violation to include a screenshot if you didn't actually submit the form!)

Thank you for your feedback once again!

Problem 3: The Complete Solution (15 pts)

Before beginning this problem, make sure you've read both **Chapter 6.3** and **Chapter 6.4**.

a) (3 pts) Consider the matrix A and vector $\vec{b}$ defined below.

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 2 & -1 \\ 1 & -1 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \end{bmatrix}$$

Find the vector $\vec{x}^*$ that minimizes $\|\vec{b} - A\vec{x}\|^2$. Show your work, but feel free to use `numpy` to compute some of the matrix operations; **here** is a video that walks through how to do this. (We're intentionally using different variables here than in the notes to have you think about the problem in general terms.)

For the remainder of the problem, we will use the same vector $\vec{b}$, but instead use the following matrix A.

$$A = \begin{bmatrix} 1 & 0 & 1 & -4 & 4 \\ 0 & 2 & 1 & -5 & 3 \\ 2 & -6 & -1 & 7 & -1 \\ 1 & -4 & -1 & 6 & -2 \end{bmatrix}$$

Note that this new matrix A has **a rank of 2**.

b) (2 pts) Now, find **one** vector $\vec{x}^*$ that minimizes $\|\vec{b} - A\vec{x}\|^2$ for this new matrix A. Again, show your work. If you've read **Chapter 6.4** closely, this should not require much calculation.

c) (4 pts) Find a basis for nullsp(A). *Hint: Try and do so efficiently, since this is the type of problem we'll see on Midterm 2.*

d) (1 pts) Show that if $\vec{x}'$ satisfies the normal equation, $A^TA\vec{x}' = A^T\vec{b}$, and $\vec{x}_0 \in$ nullsp(A), then $\vec{x}' + \vec{x}_0$ also satisfies the normal equation. *Hint: This is two-line solution; we're mostly asking it so that you interalize **what** this means and why it's true.*

e) (2 pts) Describe, using set notation, the complete set of vectors $\vec{x}^*$ that minimize $\|\vec{b} - A\vec{x}\|^2$. Is this set a subspace?

f) (3 pts) There are infinitely many vectors $\vec{x}^*$ that minimize $\|\vec{b} - A\vec{x}\|^2$. If we try and use code to find a solution, it can't return all of them — it'll pick a particular one.
In Python, use `np.linalg.lstsq` to find a vector $\vec{x}^*$ that minimizes $\|\vec{b} - A\vec{x}\|^2$. Include a screenshot of your code and the vector $\vec{x}^*$ it returns, and in your PDF, write out the coefficients of $\vec{x}^*$ as a vector (in addition to the screenshot). Then, provide an educated guess of **why** you think it picked the $\vec{x}^*$ that it did.

Problem 4: The Projection Matrix (16 pts)

The projection of a vector $\vec{y} \in \mathbb{R}^n$ onto colsp(X) is the vector $\vec{p} \in$ colsp(X) that is closest to $\vec{y}$. We can write $\vec{p} = X\vec{w}^*$, where $\vec{w}^*$ minimizes $\|\vec{y} - X\vec{w}\|^2$.

Throughout this problem, let X be an $n \times d$ matrix with linearly independent columns. Then X^TX is invertible, and

$$\vec{p} = X(X^TX)^{-1}X^T\vec{y} = P\vec{y}, \qquad P = X(X^TX)^{-1}X^T$$

The $n \times n$ matrix P is called the **projection matrix**, discussed in **Chapter 6.4**.

This is not a new idea: **Homework 5, Problem 5** was about the projection matrix for a 1-dimensional column space. Here, we're looking at the projection matrix for a column space of any dimension!

a) (7 pts) In this part only, suppose $X = \vec{x}$, where $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ is a unit vector in $\mathbb{R}^2$. Then,

$$P = \vec{x}\vec{x}^T = \begin{bmatrix} x_1^2 & x_1x_2 \\ x_1x_2 & x_2^2 \end{bmatrix}$$

Find:

(i) rank(P)

(ii) A basis for colsp(P)

(iii) A basis for nullsp(P)

(iv) A basis for colsp(P^T)

b) (4 pts) For the rest of the problem, return to assuming X is an $n \times d$ matrix with linearly independent columns, and

$$P = X(X^TX)^{-1}X^T$$

Show that P is both symmetric (meaning that $P^T = P$) and idempotent (meaning that $P^2 = P$). Then, explain in English how P's idempotence relates to projecting $\vec{y}$ onto colsp(X).

c) (2 pts) Suppose $d < n$. Explain why P is not invertible, and why the transformation $f(\vec{y}) = P\vec{y}$ can't be reversed.

d) (3 pts) Now suppose $d = n$. What is P? What does this say about the relationship between $\vec{y}$, $\vec{p}$, and colsp(X)? *Hint: Use the fact that* $(AB)^{-1} = B^{-1}A^{-1}$.

Problem 5: Same, but Different (11 pts)

In Chapter 2.4, we were introduced to one of many formulas for the optimal slope, w_1^*, and optimal intercept, w_0^*, for the simple linear regression model $h(x_i) = w_0 + w_1 x_i$ when using squared loss:

$$w_1^* = r\frac{\sigma_y}{\sigma_x} \qquad w_0^* = \bar{y} - w_1^* \bar{x}$$

The end goal of Chapters 3 through 6 has been to give us the tools to revisit the simple linear regression model in terms of linear algebra, so that we can extend our model to allow for multiple input variables. As we see in Chapter 7.1, the solution is to define the $n \times 2$ "**design matrix**" X and observation vector $\vec{y} \in \mathbb{R}^n$ as follows:

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Then, the vector containing the optimal model parameters is

$$\vec{w}^* = (X^T X)^{-1} X^T \vec{y} = \begin{bmatrix} w_0^* \\ w_1^* \end{bmatrix}$$

It's not immediately obvious why the components of $\vec{w}^*$ should have anything to do with the correlation, means of x and y, and standard deviations of x and y. In this problem, we will prove that both of these formulations are equivalent, for any dataset (x_1, y_1), (x_2, y_2), ..., (x_n, y_n).

a) (4 pts) Express the matrix $(X^T X)^{-1}$ using constants and/or summations involving x_i and/or y_i.

b) (3 pts) Prove that

$$(X^T X)^{-1} = \frac{1}{n\sigma_x^2} \begin{bmatrix} \sigma_x^2 + \bar{x}^2 & -\bar{x} \\ -\bar{x} & 1 \end{bmatrix}$$

Hint: Start by proving $\sum_{i=1}^n x_i^2 = n\sigma_x^2 + n\bar{x}^2$.

c) (4 pts) Finally, prove that

$$(X^T X)^{-1} X^T \vec{y} = \begin{bmatrix} \bar{y} - r\frac{\sigma_y}{\sigma_x}\bar{x} \\ r\frac{\sigma_y}{\sigma_x} \end{bmatrix}$$

Hint: Start by proving that $\sum_{i=1}^n x_i y_i = nr\sigma_x\sigma_y + n\bar{x}\bar{y}$.

Note that the second component of the vector above is $w_1^* = r\frac{\sigma_y}{\sigma_x}$ and the first component of the vector above is $w_0^* = \bar{y} - r\frac{\sigma_y}{\sigma_x}\bar{x} = \bar{y} - w_1^*\bar{x}$, as we first saw in Chapter 2.4! I think this is beautiful.

Problem 6: Putting it into Practice (8 pts)

This problem asks you to apply the concepts in **Chapter 7.2**, and follows the last problem.

Suppose we'd like to fit a hypothesis function of the form $h(x_i) = w_0 + w_1 x_i^2$. Notice the squared term; this is **not** a simple linear regression line.

To do so, we'll find the optimal parameter vector $\vec{w}^*$ that satisfies the normal equations. The first 5 rows of our dataset are as follows, though note that our dataset has n rows in total.

x_i	y_i
2	4
-1	2
3	5
-7	3
3	-7
$\vdots$	$\vdots$

Suppose $x_1, x_2, ..., x_n$ have a mean of $\bar{x} = 5$ and a variance of $\sigma_x^2 = 8$.

a) (2 pts) Write the first five rows of the design matrix, X.

b) (3 pts) Suppose that after solving the normal equations, we find $\vec{w}^* = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$. Find the augmented feature vector $\text{Aug}(\vec{x}_3)$ and squared loss for row 3 of the dataset.

c) (3 pts) Let $X_{\text{tri}} = 3X$, where X is the full design matrix for our dataset with n rows. Determine the bottom-left value in the matrix $X_{\text{tri}}^T X_{\text{tri}}$, i.e. the value in the second row and first column. Your answer should be an expression involving n. *Hint: You can use any of the hints or results from Problem 5 without needing to re-prove them.*

Problem 7: Orthogonalization (22 pts)

Before starting, refer to Chapter 6.5, written just for this problem. It won't be possible to do this problem without referencing it.

In parts **a)** through **d)**, we'll refer to the vectors $\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$.

a) (5 pts) By hand, apply the Gram-Schmidt process to the vectors $\vec{v}_1, \vec{v}_2, \vec{v}_3$ to find an orthonormal set of vectors $\vec{q}_1, \vec{q}_2, \vec{q}_3$. Show your work; you cannot use `numpy` for this.

Then, create the matrix $Q = \begin{bmatrix} | & | & | \\ \vec{q}_1 & \vec{q}_2 & \vec{q}_3 \\ | & | & | \end{bmatrix}$ and confirm that $Q^T Q = I$, but that $QQ^T \neq I$.

b) (2 pts) Suppose $\vec{v}_4 = \begin{bmatrix} 2 \\ 2 \\ 3 \\ 3 \end{bmatrix}$. If you were to apply the Gram-Schmidt process to the vectors $\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4$, what would the vector $\vec{Q}_4$ be? Why?

c) (3 pts) Consider $\vec{y} = \begin{bmatrix} 3 \\ 1 \\ 2 \\ 4 \end{bmatrix}$. $\vec{y}$ **is** in $\text{span}(\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}) = \text{span}(\{\vec{q}_1, \vec{q}_2, \vec{q}_3\})$.

Find scalars a, b, and c such that $a\vec{q}_1 + b\vec{q}_2 + c\vec{q}_3 = \vec{y}$, **without** solving a system of 3 equations and 3 unknowns. Instead, use the fact that $\vec{q}_1, \vec{q}_2, \vec{q}_3$ are orthonormal. *Hint: There's a relevant problem from Lab 4.*

d) (3 pts) Consider $\vec{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$. Unlike in **c)**, $\vec{y}$ **is not** in $\text{span}(\{\vec{v}_1, \vec{v}_2, \vec{v}_3\})$.

Find the vector in $\text{span}(\{\vec{v}_1, \vec{v}_2, \vec{v}_3\})$ that is closest to $\vec{y}$. **Do not** stack the $\vec{v}_i$'s into a matrix X and then use $X(X^T X)^{-1} X^T \vec{y}$. Instead, use the fact that $\vec{q}_1, \vec{q}_2, \vec{q}_3$ are orthonormal and have the same span as $\vec{v}_1, \vec{v}_2, \vec{v}_3$. How does this simplify the problem?

e) (4 pts) Open the **the supplemental Jupyter Notebook** we've created for Homework 7, which can either be found **here** in the course GitHub repository, or **here** on DataHub.

There, you're asked to implement the function `orthogonalize`, which takes in an $n \times d$ matrix V whose columns are linearly independent, and returns a matrix Q whose columns are orthonormal and have the same span as V. This problem is **not autograded**. Rather, in your submission to this part, include a screenshot of your implementation and sample output in your PDF for Homework 7.

f) (4 pts) A QR decomposition of a matrix A is a factorization of the form

$$A = QR$$

where Q is an $n \times d$ matrix with orthonormal columns and R is an $d \times d$ **upper triangular** matrix (a matrix that has 0s below the diagonal).

For example, if $A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix}$, a QR decomposition of A is

$$A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{2} \\ -1/\sqrt{3} & 2/\sqrt{6} & 0 \\ 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \end{bmatrix}}_{Q} \underbrace{\begin{bmatrix} \sqrt{3} & 2\sqrt{3}/3 & 2\sqrt{3}/3 \\ 0 & \sqrt{6}/3 & 5\sqrt{6}/6 \\ 0 & 0 & 1/\sqrt{2} \end{bmatrix}}_{R}$$

Finding the Q in a QR decomposition is straightforward: apply Gram-Schmidt to the columns of A, assuming A's columns are linearly independent. The question is how to find R.

(i) In the supplemental Jupyter Notebook, we've defined an arbitrary matrix A and call your `orthogonalize` function on it, and give you hints as to how to find R. Using the experimentation there, and what you know about Q, **explain how to find R.**

(ii) Find a QR decomposition of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. Note that the columns of A are made up of the same three vectors you worked with in parts **a)** through **d)** of this problem.

(iii) Given **a** QR decomposition of A, explain how to find **another** QR decomposition of A with a (slightly) different Q and/or R.

g) (1 pt) Suppose we replace the design matrix X with a matrix Q whose columns are orthonormal. How does the normal equation $X^T X \vec{w}^* = X^T \vec{y}$ simplify? Give an expression for $\vec{w}^*$.

Problem 8: Billy the Waiter (13 pts)

This problem involves writing code and submitting it to the Pensive autograder. The goal of this problem is to give you a taste of how linear algebra can be used to implement linear regression in code, and show you how to build models that involve multiple features (including categorical variables).

Open the **the supplemental Jupyter Notebook** we've created for Homework 7, which can either be found **here** in the course GitHub repository, or **here** on DataHub.

This problem is entirely autograded; to receive credit for Problem 8 of this homework, you'll need to submit your completed notebook to the autograder on Pensive. Your submission time for Homework 7 is the **latest** of your video link, PDF, and code submission times.